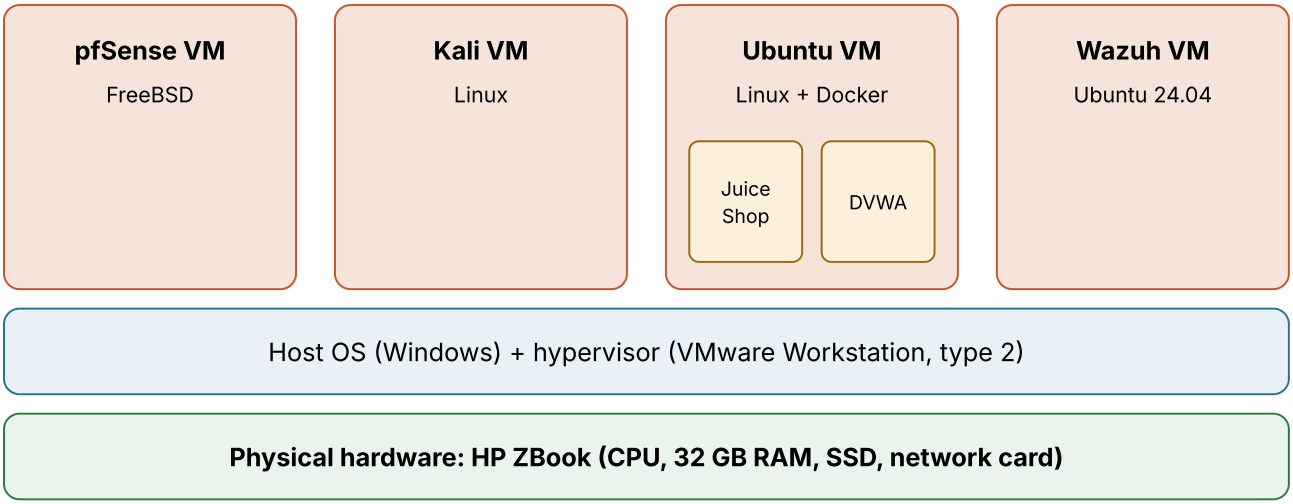
CLOUD SECURITY PATH · STUDY GUIDE 2

Virtualization & Containers

The foundation of cloud computing, explained with the lab you already run. Hypervisors, virtual hardware and networking, snapshots and backups, Docker, container security and Kubernetes basics. Each chapter ends with a quick check.



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1 What virtualization is

> **Virtualization uses software to split one physical computer into several virtual machines (VMs). Each VM behaves like a complete computer with its own operating system.**

Before virtualization, one server ran one operating system, and most servers sat 80–90% idle. Virtualization lets one physical server run many VMs, so hardware is used efficiently, and machines can be created, copied and deleted in minutes instead of weeks.

Term	Meaning	In your lab
Host	The physical machine	Your HP ZBook
Guest	A VM running on the host	pfSense, Kali, Ubuntu, Wazuh
Hypervisor	The software that creates and runs VMs	VMware Workstation Pro
Guest OS	The operating system inside a VM	FreeBSD (pfSense), Kali Linux, Ubuntu

THINK OF IT LIKE THIS

The host is an apartment building. The hypervisor is the building manager, who gives each flat (VM) its own walls, door, electricity meter and water supply from the building's shared resources. Tenants (guest operating systems) live independently and shouldn't be able to walk into each other's flats.

Why it matters for your career

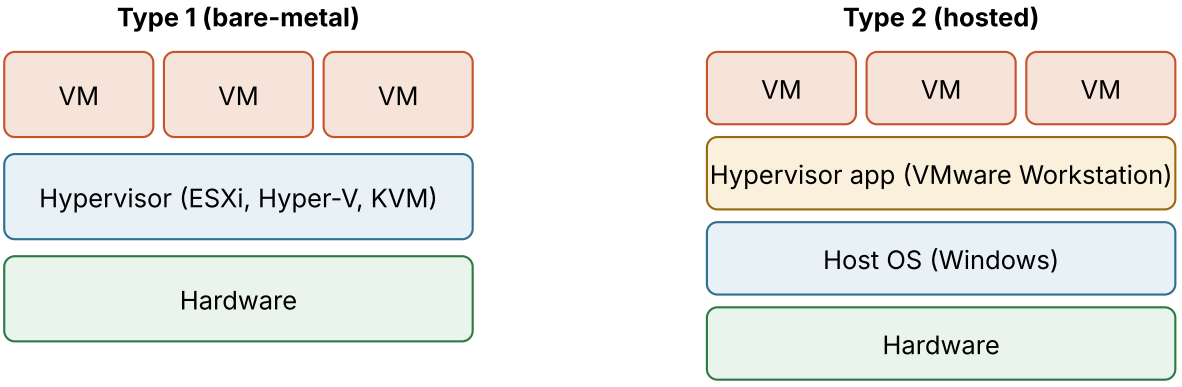
- **The cloud is virtualization at massive scale.** When you start an Azure VM or AWS EC2 instance, a hypervisor in a data centre creates it for you.
- Labs, malware analysis sandboxes and SOC tooling all run on VMs.
- For a small business security service, a few VMs on one server can host the monitoring stack (for example Wazuh).

QUICK CHECK (answers at the back)

1. What is the difference between the host and a guest?
2. Why did virtualization save companies money?

2 Hypervisors: type 1 and type 2

> **A type 1 hypervisor runs directly on the hardware. A type 2 hypervisor runs as an application on top of a normal operating system.**



	Type 1	Type 2
Examples	VMware ESXi, Microsoft Hyper-V, KVM, Xen, Proxmox VE	VMware Workstation, VirtualBox, Parallels
Used for	Data centres and **all major clouds**	Labs, development, desktops
Performance	Higher: nothing between the VMs and the hardware	Lower: the host OS uses some resources
Attack surface	Smaller: a minimal hypervisor	Larger: a whole desktop OS underneath

The clouds: Azure runs on a customised version of **Hyper-V**. AWS's **Nitro** system uses a lightweight hypervisor based on **KVM**, with much of the work moved onto dedicated hardware cards. Google Cloud uses KVM.

Hardware support

Modern CPUs include virtualization features (**Intel VT-x**, **AMD-V**) that let VMs run at near-native speed. They're switched on in the BIOS or UEFI, and VMware needs them. **Nested virtualization** means running a hypervisor inside a VM, which some cloud VM sizes support.

REMEMBER

Type 1 runs on the metal; type 2 runs on an OS. The cloud is type 1. Your laptop lab is type 2. The concepts are identical.

QUICK CHECK (answers at the back)

3. Name one type 1 and one type 2 hypervisor.
4. Which hypervisor family does Azure use, and which does AWS Nitro use?
5. Why does a type 1 hypervisor have a smaller attack surface?

3 Virtual hardware

> **The hypervisor gives each VM virtual CPUs, memory, disks and network cards, which are slices of the real hardware.**

Resource	How it works	Watch out for
vCPU	A share of a real CPU core, scheduled by the hypervisor	**Overcommitting**: more vCPUs than real cores makes VMs wait, as when your Wazuh boot timed out
Memory	A block of real RAM reserved for the VM	The host needs some RAM too. Running VMs past the total makes everything swap to disk and crawl.
Virtual disk	A file on the host (`.vmdk` in VMware; VHD/VHDX in Hyper-V and Azure)	**Thin** disks grow as used; **thick** disks reserve all space up front. A slow external SSD slows every VM on it.
Virtual NIC	A software network card with its own MAC address	Which virtual network it's attached to decides who it can reach (chapter 4)

Your lab's resource budget (32 GB)

VM	RAM	vCPU
pfSense	1–2 GB	1–2
Kali	4 GB	2
Ubuntu DMZ	2–4 GB	2
Wazuh	8 GB	4
Windows Server + Windows 11 (next)	4 GB each	2 each

In the cloud: VM sizes

Clouds sell fixed combinations of vCPU and RAM called **sizes** (Azure, e.g. `B2s`) or **instance types** (AWS, e.g. `t3.small`). You pay per second or hour while the VM runs, so **stopping or deleting VMs you don't need** is both a cost and a security habit.

IN YOUR LAB

Open **VM → Settings** for Wazuh in VMware and read its memory, processors, disk and network adapter. You're looking at exactly the settings an Azure VM size describes.

QUICK CHECK (answers at the back)

6. What happens if you give VMs more RAM in total than the host has?
7. Thin vs thick disk: which saves space?

4 Virtual networking

> **Hypervisors build networks in software: virtual switches connect VMs to each other, to the host, or to the outside world.**

VMware network type	What it does	Cloud equivalent
NAT	The VM shares the host's internet connection through a virtual router	A NAT gateway
Bridged	The VM appears directly on your real home network, with its own IP from your router	A public IP on the VM (avoid)
Host-only	The VM can talk only to the host and other host-only VMs	A subnet with no internet route
LAN segment (your lab)	A private virtual switch: only VMs attached to it can talk	A **subnet** in a VNet/VPC

Your pfSense VM has **four virtual NICs**: one on NAT (WAN) and one on each LAN segment. It routes and filters between them, which is exactly what cloud routing plus network security groups do.

Your lab

- LAN segments (soc-lab, soc-dmz, soc-mgmt)
- pfSense routing between segments
- pfSense firewall rules
- VMware NAT to the internet

Azure / AWS

- Subnets inside a VNet / VPC
- Route tables (automatic between subnets)
- NSGs / security groups
- NAT gateway

REMEMBER

One big difference: in the cloud, subnets in the same VNet can reach each other **by default**. Segmentation only exists if you add NSGs or security groups. In your lab, separate segments are isolated until pfSense allows traffic.

QUICK CHECK (answers at the back)

8. Which VMware network type is closest to a cloud subnet?
9. Why is a bridged VM riskier than one on a LAN segment?

5 The console and remote management

> **The console is a VM's "physical" screen and keyboard. It works even when the network is broken, which is why admins rely on it for recovery.**

- **VMware console:** the window where you typed on Wazuh. It's a virtual monitor and keyboard, with no network involved. That's why it worked when SSH was blocked, and why copy and paste didn't work until you used SSH from Kali.
- **Remote access over the network:** SSH (Linux), RDP (Windows), web GUIs (pfSense, Wazuh). These are faster and support copy and paste, but they depend on networking and firewall rules.
- **In the cloud:** the **Azure Serial Console** and **AWS EC2 Serial Console** are the equivalent of the VMware console, for when networking is broken. **Azure Bastion**, **AWS Systems Manager Session Manager** and **EC2 Instance Connect** give SSH/RDP access **without a public IP**.
- **Management planes:** the Azure portal, AWS console and their command-line tools (`az`, `aws`) create and control VMs. Whoever controls these accounts controls every VM. Protect them with MFA and least privilege.

REMEMBER

Out-of-band access (the console) is for recovery. Day-to-day access goes through hardened, logged paths: SSH keys, Bastion, Session Manager. Never put SSH or RDP directly on the internet.

QUICK CHECK (answers at the back)

10. Why could you type on the Wazuh console when SSH from Kali wasn't allowed?
11. Name one way to reach a cloud VM without giving it a public IP.

6 Snapshots, images and backups

> **A snapshot freezes a VM's state so you can roll back. A backup is a separate, independent copy you can restore even if the original is gone. They are not the same thing.**

	Snapshot	Backup	Image / template
What	A point-in-time state of a VM's disks (and optionally memory)	An independent copy stored elsewhere	A master copy used to create new VMs
Stored	Usually alongside the VM	**Somewhere else**: another disk, site or cloud	An image library or gallery
Protects against	A bad change ("undo")	Hardware loss, ransomware, deletion	Inconsistent builds
In your lab	`wazuh-02-first-detection` etc.	Not yet (Phase 6)	The Ubuntu ISO install
Cloud	Managed disk snapshots / EBS snapshots	Azure Backup / AWS Backup	Azure Compute Gallery / AMIs

The 3-2-1 backup rule

3 copies of your data, on **2** different types of storage, with **1** copy off-site. Modern advice adds: **1 copy offline or immutable** (it can't be changed or deleted, even by an admin), and **0 errors** on a tested restore.

REMEMBER

A backup you've never restored is a hope, not a backup. Test restores on a schedule. For a small business security service, a tested backup is often the single most valuable control against ransomware.

Security of snapshots

- A snapshot contains a **full copy of the disk**, including passwords, keys and data. A leaked or publicly shared cloud snapshot is a data breach.
- Memory snapshots can contain secrets that were only ever held in RAM.
- In incident response, snapshots are used to **preserve evidence** before cleaning up a compromised VM.

QUICK CHECK (answers at the back)

12. Why is a snapshot not a backup?
13. Explain the 3-2-1 rule.
14. How are snapshots used in incident response?

7 Virtualization security

> **The hypervisor is the foundation everything else trusts. If it's compromised, every VM on it is compromised.**

Risk	What it is	Control
VM escape	Code in a VM breaks out into the hypervisor or other VMs. Rare, but critical.	Patch hypervisors promptly; remove unneeded virtual devices (USB, shared folders)
Weak isolation settings	Shared folders, clipboard and drag-and-drop can carry malware between guest and host	Disable sharing for malware or attack VMs
Management plane compromise	Stolen admin access to vCenter, Azure or AWS gives control of every VM	MFA, least privilege, separate admin accounts, logging
VM sprawl	Forgotten, unpatched VMs that nobody owns	Inventory, tags, owners, automatic shutdown
Snapshot and image leakage	Copies of disks shared or exposed	Restrict sharing, encrypt, review public exposure
Noisy neighbour / side channels	One VM affects or observes another on the same host	Cloud providers mitigate this; dedicated hosts exist for sensitive workloads

IN YOUR LAB

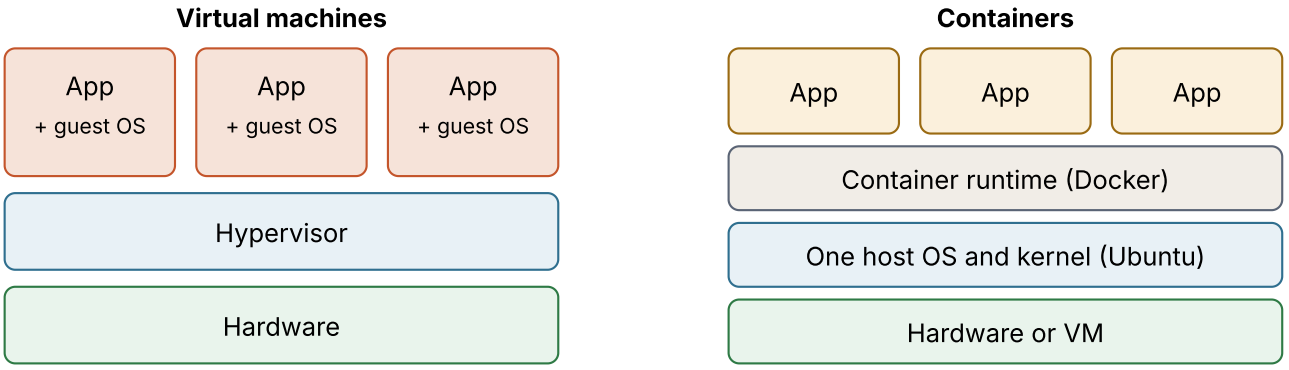
Your attacker (Kali) and targets (Ubuntu) are VMs on the same host. That's safe because they're isolated by LAN segments and pfSense, and the vulnerable apps never touch a bridged network.

QUICK CHECK (answers at the back)

15. What is VM escape, and what is the main defence?
16. Why is the cloud management plane such a valuable target?

8 Containers

> **A container packages an application with everything it needs, and runs it as an isolated process that shares the host's kernel. It's lighter and faster than a VM.**



	Virtual machine	Container
Isolates	A whole OS, with its own kernel	A process, sharing the host kernel
Size and start time	Gigabytes; minutes	Megabytes; seconds
Isolation strength	Stronger (hardware-level)	Weaker (kernel-level). A kernel bug can affect all containers.
Best for	Different OSes, strong isolation	Packaging and scaling applications

Key container terms

- **Image:** a read-only template (app plus libraries), built in **layers**. Example: `bkimminich/juice-shop`.
- **Container:** a running instance of an image.
- **Registry:** where images are stored and pulled from (Docker Hub, Azure Container Registry, Amazon ECR).
- **Dockerfile:** the recipe for building an image.
- **Volume:** storage that survives when a container is deleted.

How Linux isolates containers

Namespaces give each container its own view of processes, network, files and users. **Cgroups** (control groups) limit how much CPU and memory each container can use. Both are Linux kernel features, which is why containers are a Linux-first technology. That's another reason Linux skills matter for cloud.

QUICK CHECK (answers at the back)

17. What does a container share with its host that a VM does not?
18. Image vs container: what's the difference?
19. What do namespaces and cgroups do?

9 Docker in your lab

> **Your Ubuntu DMZ server already runs three containers: Juice Shop, DVWA and DVWA's MariaDB database.**

Command	What it does
`docker ps` / `docker ps -a`	Running containers / all containers, including stopped ones
`docker images`	Images stored locally
`docker logs <name>`	A container's output, often its application log
`docker exec -it <name> sh`	Open a shell inside a running container
`docker inspect <name>`	Full details: image, mounts, network, user, ports
`docker stop` / `start` / `rm`	Stop, start or delete a container
`docker compose up -d` / `down`	Start or stop a multi-container app defined in `compose.yml`
`docker network ls`	Docker's own virtual networks

Port publishing

`-p 3000:3000` means "connect port 3000 on the host to port 3000 in the container". `127.0.0.1:4280:80` publishes only to the host itself, which is why DVWA was unreachable from Kali until you changed it.

REMEMBER

Docker writes its own firewall rules, so **published ports bypass ufw**. On your server, pfSense is the real control for the web apps. In the cloud, the equivalent control is the NSG or security group in front of the VM.

QUICK CHECK (answers at the back)

20. Which command opens a shell inside a running container?

21. What does `-p 8080:80` mean?

10 Container security

> **Containers move fast, so security has to be built into how images are made, stored and run. This is the start of DevSecOps.**

Stage	Risk	Controls
Build	Vulnerable or malicious packages in the image; secrets baked in	Official, minimal base images; **image scanning** (e.g. Trivy); never put passwords or keys in a Dockerfile
Store	Tampered or public images	A private registry; signed images; pull only from trusted sources
Run	Containers running as root; too many privileges; open ports	Run as a non-root user; no `--privileged`; drop Linux capabilities; read-only filesystem; CPU and memory limits
Host	The Docker socket (`/var/run/docker.sock`) gives root-level control of the host	Never mount the socket into containers; limit membership of the `docker` group; patch the host kernel
Network	Published ports bypassing the host firewall	Publish only what's needed; bind to specific IPs; network firewall in front
Monitor	Attacks inside containers go unseen	Collect container logs; runtime detection; Wazuh's Docker monitoring

REMEMBER

Being in the `docker` group is effectively root on the host. You added your Ubuntu user to it for convenience, which is fine in a lab, but treat it as a privileged group in production.

QUICK CHECK (answers at the back)

22. Why is running a container as root risky?
23. Why should the Docker socket never be mounted into a container?
24. What does an image scanner do?

11 Kubernetes basics

> **Kubernetes (K8s) runs and manages many containers across many machines: it restarts failed containers, scales them up and down, and routes traffic to them.**

Term	Meaning
Cluster	A set of machines (nodes) managed together
Node	A VM or server that runs containers
Pod	The smallest unit: one or more containers deployed together
Deployment	Says "keep N copies of this pod running"
Service	A stable network address in front of a group of pods
Namespace	A logical partition of the cluster, e.g. per team
Control plane / API server	The brain of the cluster. Whoever controls it controls everything.

Managed Kubernetes: AKS (Azure), **EKS** (AWS) and GKE (Google) run the control plane for you, which is part of the shared responsibility model (Guide 3).

Kubernetes security essentials

- **RBAC:** least-privilege access to the API server.
- **Network policies:** by default all pods can talk to each other. Policies add micro-segmentation.
- **Secrets management:** store secrets in a vault (Azure Key Vault, AWS Secrets Manager), not in plain config.
- **Pod security:** no privileged pods, non-root users, read-only filesystems.
- **Audit logs:** the API server's audit log is the key detection source.

IN YOUR LAB

Phase 7 adds **k3s**, a lightweight Kubernetes, on the Ubuntu server, so you can practise these controls without cloud costs.

QUICK CHECK (answers at the back)

25. What is a pod?
26. By default, can every pod talk to every other pod? What fixes that?

12 From your lab to the cloud

> **Every piece of your home lab has a cloud equivalent. Learning it locally first makes the cloud versions familiar.**

Home lab	Azure	AWS
VMware VM	Azure Virtual Machine	EC2 instance
VM settings (RAM, vCPU)	VM size (e.g. B2s)	Instance type (e.g. t3.small)
Virtual disk (.vmdk)	Managed disk	EBS volume
VMware snapshot	Disk snapshot	EBS snapshot
Ubuntu ISO install	Marketplace image	AMI
LAN segment	Subnet in a VNet	Subnet in a VPC
pfSense rules	NSG	Security group / NACL
VMware console	Serial Console	EC2 Serial Console
SSH from Kali	Azure Bastion	Session Manager
Docker on Ubuntu	Container Instances / App Service	ECS / Fargate
Docker Hub image	Azure Container Registry	Amazon ECR
k3s (Phase 7)	AKS	EKS
Backups (Phase 6)	Azure Backup	AWS Backup

QUICK CHECK (answers at the back)

27. What are the Azure and AWS names for a VMware virtual disk?
28. What replaces "SSH from Kali" in a secure cloud design?

13 Glossary

Term	Meaning
Bare-metal	Running directly on hardware, with no OS underneath
Cgroups	Linux feature that limits a container's CPU and memory
Container	An isolated process running from an image, sharing the host kernel
Guest / host	The VM / the physical machine it runs on
Hypervisor	Software that creates and runs VMs (type 1 or type 2)
Image	A read-only template for containers or VMs
Immutable backup	A backup that can't be altered or deleted during its retention period
Namespace (Linux)	Gives a container its own view of processes, network and files
Overcommit	Allocating more virtual resources than physically exist
Pod	The smallest deployable unit in Kubernetes
Registry	A store of container images
Snapshot	A point-in-time state of a VM's disks
VM escape	Breaking out of a VM into the hypervisor
vCPU	A virtual CPU, scheduled onto real cores

Answers

1. The host is the physical machine; a guest is a VM running on it.
2. One server could run many workloads, so idle hardware was used, and machines could be created in minutes.
3. Type 1: ESXi, Hyper-V, KVM, Xen or Proxmox. Type 2: VMware Workstation, VirtualBox.
4. Azure: Hyper-V. AWS Nitro: a lightweight KVM-based hypervisor.
5. It's a minimal system with no full desktop OS underneath, so there's less code to attack.
6. The host starts swapping to disk and every VM slows down dramatically.
7. Thin.
8. LAN segment.
9. It sits directly on your real network, reachable by other devices, instead of behind a firewall.
10. The console is a virtual screen and keyboard; it doesn't use the network.
11. Azure Bastion, AWS Session Manager or EC2 Instance Connect.
12. It's stored with the VM and depends on it. If the VM's storage is lost or encrypted by ransomware, the snapshot goes too.
13. 3 copies, 2 types of storage, 1 off-site (plus ideally 1 offline or immutable, and tested restores).
14. To preserve the state of a compromised machine as evidence before cleaning it up.
15. Code escaping from a VM into the hypervisor; patch the hypervisor promptly.
16. It controls every VM and resource, so one stolen account can compromise everything.
17. The kernel.
18. An image is the template; a container is a running instance of it.

19. Namespaces isolate what a container can see; cgroups limit what it can use.
20. `docker exec -it <name> sh`
21. Host port 8080 is forwarded to port 80 inside the container.
22. If an attacker breaks out of the container, they may land as root on the host.
23. Access to the socket means full control of Docker, and therefore root on the host.
24. It lists known vulnerabilities in the packages inside an image.
25. One or more containers deployed together as a unit.
26. Yes; network policies restrict it.
27. Azure: managed disk. AWS: EBS volume.
28. Azure Bastion or AWS Session Manager, with no public IP on the VM.